EnAbled Power

Early warning and risk assessment tools to help medically at-risk populations and TX regulators manage power outages using calculus and statistical approaches

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MOTIVATION

- Extended power **outages** pose a serious **life-threatening** risk to medically vulnerable populations
- **Government** support systems often become **overwhelmed** during prolonged outages [1],[2].
- Proposed **state-funded programs** such as home generators for vulnerable populations will require **prioritization** due to budget constraints.

Figure 1. Impact of 2021 Winter Storm Uri
Image taken from https://www.click2houston.com

RESEARCH QUESTIONS

- How do we **prioritize Texas counties** for state-level programs such as home generator provisioning ?
- Since prolonged outages usually develop over 24 hours, **is it possible to give an early warning** to medically vulnerable people and local resources ?

RESEARCH OBJECTIVES

- Develop **data-driven analysis** to *classify and quantify outage* likelihood from **multi-year** data.
- Develop ***risk analysis*** to **rank Texas counties** with medically-at-risk population and high risk of prolonged outages.
- Develop **early warning** of extended outages

PREVIOUS WORK

- Previous work of power outage analysis [6] **focused on specific major events** such as hurricanes, winter storms **instead of multi-year trends**.
- Analysis and recommendations from previous work has not been used to provide **direct recommendations** for medically-at risk individuals

METHODOLOGY

Collect publicly available data

254 Texas counties, multi-year outage at 15 min intervals from [3] and medically at-risk populations [5]

COUNTY LEVEL ANALYSIS

OUTAGE EVENT RECOGNITION
Start, end, duration

→ **CLASSIFY EVENTS and LIKELIHOOD**
< 24, 24-48, 48-72 hr

→ **COMPUTE RISK FACTOR**
Likelihood * impact

COMPUTE INFLECTION POINT
Early indicator prior to peak outage

STATE LEVEL ANALYSIS

Rank Texas counties based on RISK FACTOR analysis to prioritize state-funded programs

MAIN RESULTS & CONCLUSIONS

COUNTY LEVEL OUTAGE TYPE ANALYSIS

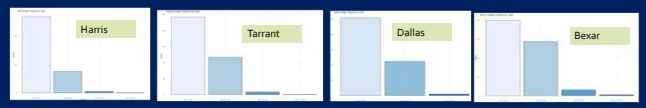

Figure 2. Classifications of outage events at Texas counties with highest medically at-risk population *generated by I.B. Sinaga using R-studio, 2025-2026*

STATE LEVEL RISK ANALYSIS for PRIORITIZATION

- Home generators will be critical for outage events >= 24 hrs.
- High impact outages in Texas lasted 2-3 days (Type 2 and Type 3) based on multi-year analysis

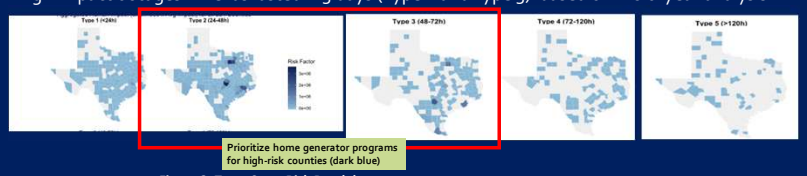

Figure 3. Texas State Risk Breakdown *generated by I.B. Sinaga using R-studio, 2025-2026*

EARLY WARNING FOR MEDICALLY AT-RISK POPULATION

- Inflection point analysis during major outage showed that developing outages can be predicted HOURS AHEAD
- The inflection Ct_max of 10,000+ customers aligns with US Department of Energy major outage category [6].

- Inflection point Ct_max
- If Ct_max >= real time outage tracker, prolonged outage is likely underway

	Ct_max	fit	d1	d2	inflection_time	county	max_outage_time	time_diff_hours
1	11356.4223	8.479369	0.247967033	3.528825e-04	2021-02-16 00:45:00	Harris	2021-02-16 08:00:00	-7.25
2	1249.2301	38.147640	-4.324863044	-2.404533e+01	2021-02-13 23:15:00	Tarrant	2021-02-15 16:00:00	-40.75
3	26039.3619	29.161469	0.036816640	1.308940e-05	2021-02-15 08:30:00	Tarrant	2021-02-15 16:00:00	-7.50
4	59953.2364	29.111498	-0.006621079	-2.431431e-08	2021-02-15 09:00:00	Dallas	2021-02-15 12:00:00	-3.00
5	839.9328	49.070129	23.504480630	2.541789e+01	2021-02-11 20:30:00	Bexar	2021-02-15 16:00:00	-91.50
6	13490.5432	75.393675	-0.140520719	-1.710211e-04	2021-02-15 07:45:00	Bexar	2021-02-15 16:00:00	-8.25

- Inflection time before peak outage occurs

Table 1. Early warning based on inflection point analysis, generated by I.B. Sinaga using R-studio, 2026

LARGE SCALE DATA PROCESSING

Using calculus and statistical approaches

OUTAGE EVENT RECOGNITION

- In [6], outage critical time stamps are characterized as **Tstart, Tpeak, Tend**. The method in [6] is heuristic and focused on specific events.
- To automatically compute these parameters from large-scale data, we proposed a calculus approach coded in R studio [8]. To filter noise, the analysis was done **every 4-hr window** as suggested in [6].
  - Affected customer over time t can be represented as a **time-based function** $C(t)$
  - **Tpeak**: when $dC(t)/dt$ changes from positive to negative to indicate (local) maxima with peak customer Cp=C($Tpeak$).
  - **Tstart**: $\frac{dC(t)}{dt}$ >0 as outage grows. Due to noise, we used Tstart as the time when when $C(t) = 0.1\ Cp$ as suggested in [6].
  - **Tend**:$dC(t)/dt$ <0 since outage drops, but due to noise this is not a reliable indicator. The drop from peak to end typically follows an **exponential decay** function, so we set Tend is the time when C(t) =**exp(-4)*** Cp [9].
  - Outage Duration D(t) = **Tstart-Tend**

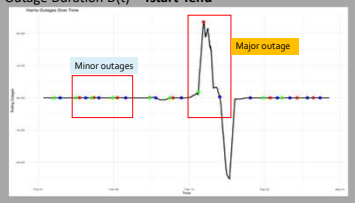

Figure.4. Outage events recognition results Feb 2021 Harris County with Tstart (green), Tpeak (red), Tend (blue). Generated by I.B. Sinaga using R-studio, 2025

OUTAGE EVENT CLASSIFICATION, LIKELIHOOD, RISK

- Outage events classified [6]: based on duration Type 1 <24 hrs, Type 2 (24-48 hrs), Type 3 (48-72 hrs), Type 4 (72-120 hrs), Type 5 (>120 hrs).
- Compute:
  - Likelihood = (Number of each Type)/Total outage
  - Impact = average customer impacted for each type
  - Risk factor = Likelihood * impact

INFLECTION POINT ANALYSIS

- Inflection points in statistics indicate the critical point of change over a period of interest, computed by R-studio [8]. Winter Storm Uri 2021 is used as period of interest in this project.
- Inflection point before the peak : an early indicator of prolonged outage
- Inflection point after the peak: indicator of restoration time

FUTURE WORK

- Create **an app of early warning** system based on inflection point analysis
- **Expand inflection point** analysis to **multiple major outages** in Texas to improve its reliability
- **Replicate** analysis to other US states

ACKNOWLEDGMENT

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REFERENCES

[1]. Tran, T. (2023, February 14). Power Outages in Texas Need Not Turn Deadly. American Association of Retired Persons. https://states.aarp.org/texas/power-outages-in-texas-need-not-turn-deadly.
[2]. Frederick, A. (2025). Disabled Power: A Storm, a grid, and Embodied Harm in the Age of Disaster. New York University Press.
[3]. Eagle-I Power Outage Data Information. (n.d.). EAGLE-I Power Outage Data Information. Constellation. https://doi.ccs.ornl.gov/dataset/d26451af-7ea4-577c-8285-78b72cd8a2dc
[4]. U.S. Department of Health and Human Services, Administration for Strategic Preparedness and Response. (2025).
[5]. HHS emPOWER Map. HHS emPOWER Program. https://empowerprogram.hhs.gov/empowermap
[6]. Ericson, S., Cox, J., Abdelmalak, M., Rabinowitz, H., & Hotchkiss, E. (2023). Exceedance Probabilities and Recurrence Intervals for Extended Power Outages in the United States (Technical Report No. NREL/TP-6A20-83092). National Renewable Energy Laboratory.
[7].Larson, R., & Edwards, B. H. (2023). Calculus of a Single Variable (12th ed.). Cengage Learning.
[8].HarvardX. (2025). Data Science Professional Certificate. edX. https://www.edx.org/certificates/professional-certificate/harvardx-data-science
[9] MIT. (2017). Exponential growth and decay. https://web.mit.edu/18.06/www/Spring17/ODEs.pdf